RANSHIELD

AI- Powered Android Ransomware Defense

Team Name- Aegis

Team Lead Name- Mr. Rishikesh Vishwas Tapkir

Team Lead University- National Forensic Sciences University, Goa

Problem Statement Domain- Mobile Device Security

Problem Statement- Ransomware Early Warning System for Android Devices

Solution Subtitle- “Ranshield” Android Mobile Application for Detection & Mitigation of Mobile Ransomwares with Data Recovery

इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY
सत्यमेव जयते
ISEA
DSCI PROMOTING DATA PROTECTION

THE PROBLEM

Mobile ransomware is the fastest-growing cyber threat — and Android is the primary target

$1.1B+

Ransomware damages paid in 2023 globally

73%

Of mobile malware targets Android devices

Zero-Day

Attacks evade traditional antivirus software

- Existing antivirus apps are REACTIVE — they detect AFTER damage is done
- No recovery mechanism once files are encrypted by ransomware
- Enterprise MDM tools don't protect personal Android devices of employees
- Sideloaded APKs bypass Play Protect entirely — the #1 infection vector

INTRODUCING RANSHIELD

The 3-layer, real-time Android ransomware defense system with built-in recovery

DETECT

Static Analysis

APK scanning via MobSF backend + XGBoost ML model analyzing 10-dimensional behavioral feature vectors in real time

BLOCK

Zero-Day Behavioral Engine

Accessibility service monitors screen, file system, and app behavior. Triggers overlay lockdown within milliseconds of ransomware activity

RECOVER

Snapshot Restore Engine

V1 data snapshots of DCIM, Documents, Downloads & Music backed up to sandboxed storage. One-click full recovery after an attack

✦ *All three layers operate simultaneously — detection triggers blocking, blocking triggers snapshot — with zero user friction*

HOW IT WORKS

A multi-stage defense architecture running entirely on-device

App Install Detected

APK Upload & MobSF Scan

XGBoost ML Prediction

Behavioral Monitoring

Threat Lockdown

Stage 2 — ML Engine Details

- 10-dimensional feature vector (6 static permissions + 4 dynamic behaviors)
- XGBoost model compiled natively to Java (RansomwarePredictor.java)
- 100-tree ensemble — <5ms inference on-device
- Strike threshold system: 1–3 strikes before lockdown based on confidence

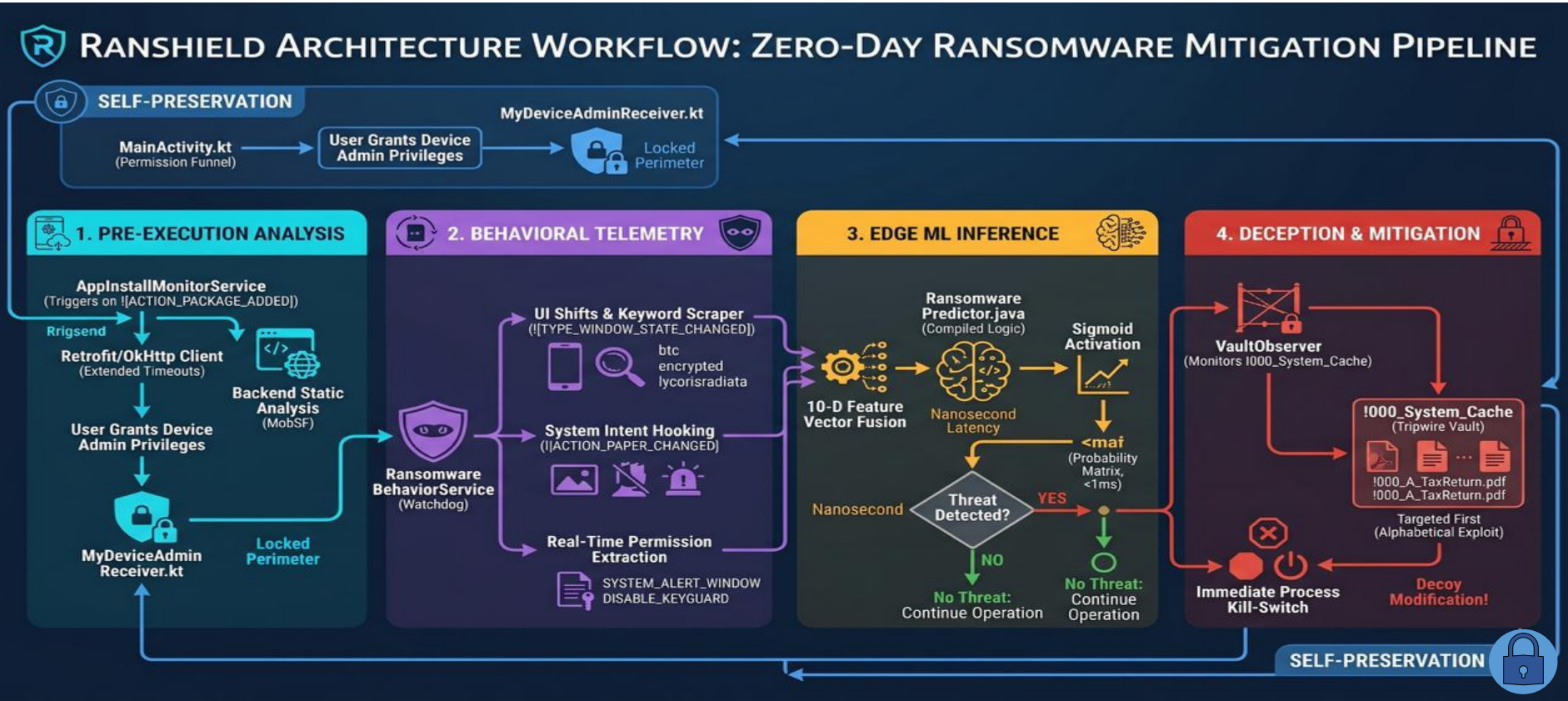
Stage 2 — Zero-Day Heuristics

- Bait/tripwire files planted in DCIM, Docs, Downloads
- FileObserver monitors MODIFY/DELETE events in real time
- Ransom keyword scanning in 2 languages (English + Chinese)
- Wallpaper change detection as ransomware behavioral signal

Stage 3 — Recovery Engine

- Full ZIP snapshot of user data before exposure
- Stored in app sandbox — inaccessible to other apps
- One-click restore overwrites encrypted ransomware files
- Forensic-grade recovery with boot-time tripwire re-planting

HOW IT WORKS- THE ARCHITECTURE



TECHNOLOGY STACK

Enterprise-grade components, fully on-device inference — no cloud dependency for detection

ON-DEVICE (Android)

Language:	Kotlin + Java (Android SDK 34)
ML Model:	XGBoost → compiled Java (RansomwarePredictor)
Monitoring:	AccessibilityService + FileObserver (VaultObserver)
Networking:	Retrofit2 + OkHttp3 (APK upload)
Serialization:	Gson (JSON response parsing)
Recovery:	ZipOutputStream — sandboxed snapshot storage
Admin:	DevicePolicyManager (device admin lock)
Overlay:	TYPE_APPLICATION_OVERLAY (instant UI lockdown)

CLOUD BACKEND

Scanner:	MobSF (Mobile Security Framework)
API Server:	Python Flask (REST endpoint)
Tunneling:	ngrok (dev) → production cloud deploy
Analysis:	Static APK analysis pipeline
Scoring:	Security score 0–100 + risk classification
Transport:	Multipart file upload (5-min timeout handling)

Future roadmap: Replace MobSF cloud with federated on-device ML training — zero cloud dependency, full privacy

MARKET OPPORTUNITY

Introducing newer security concepts in Android market

1 EDGE INFERENCE

Shifting the focus from cloud-based telemetry and analysis to on-device live telemetry monitoring, with the intelligence of ML model (XGBoost)

2 INTRODUCING DECEPTION TECHNOLOGY IN MOBILE

Deception technology like ‘tripwires’ or ‘honeypots’ are absent from Mobile environments due to heavy file sandboxing

3 INTEGRATING WITH EXISTING MDM SOLUTIONS

This ‘behavioral engine’ can be integrated with existing Mobile Device Management (MDM) solutions in Enterprise environments that follow “Bring-Your-Own-Device” policy.

WHO WE PROTECT

Ranshield can serve multiple customer tiers — from individuals to enterprise deployments

B2C — Individuals

- Android users who sideload apps (300M+)
- Users in markets with low Play Protect coverage
- Students & professionals with sensitive mobile data

Freemium → ₹199/mo Premium

B2B — SMEs & Enterprises

- BYOD workforce: employee Android devices
- IT teams needing centralized threat dashboards
- Financial, healthcare & legal sector compliance

₹999/device/yr Enterprise License

B2G — Government & Defense

- Government mobile fleets running Android
- Critical infrastructure protection requirements
- CERT-In mandated endpoint security compliance

Custom Contract

COMPETITIVE ADVANTAGE

Ranshield vs. existing mobile security solutions

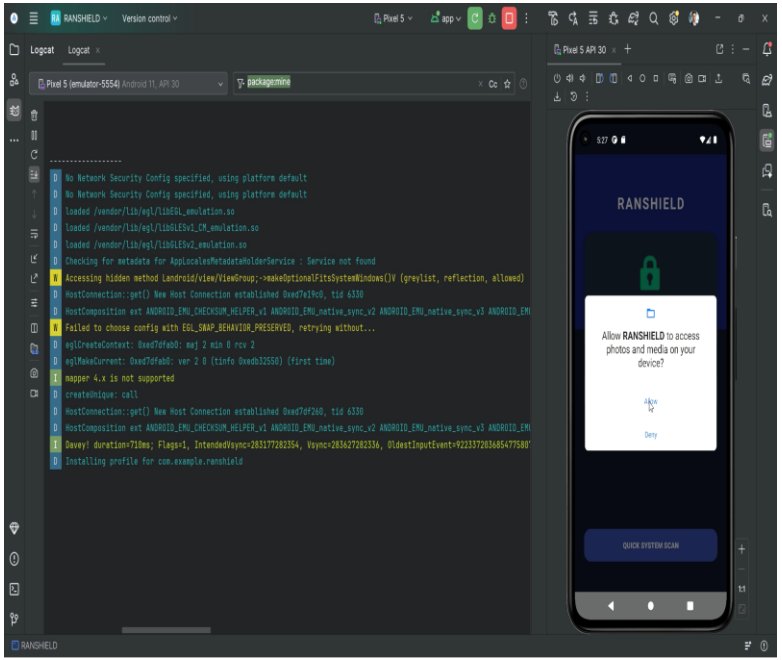
Feature	Traditional Antivirus	Google Play Protect	MDM Solutions	RANSHIELD
Real-Time Behavioral AI	✗	Partial	✗	✓
Zero-Day Ransomware Block	✗	✗	✗	✓
Data Snapshot & Recovery	✗	✗	✗	✓
Works on Sideloaded APKs	Partial	✗	✗	✓
On-Device ML (No Cloud)	✗	Partial	✗	✓
File Tripwire Detection	✗	✗	✗	✓
Overlay Lockdown UI	✗	✗	Partial	✓
Free for Individuals	✗	✓	✗	✓

Ranshield is the only solution that combines proactive ML detection + behavioral blocking + one-click recovery in a single app

TRACTION & DEMO

Proof of concept validated — working MVP demonstrated on real Android device

- Fully working Android MVP deployed on physical device
- XGBoost ML model trained, compiled & running natively in Java
- MobSF backend integrated — real APK scanning pipeline functional
- Live demo: ransomware APK detected, blocked & data recovered
- 3-layer defense (APK scan → behavioral ML → file tripwire) active
- Accessibility service, overlay UI & device admin all functional



Ransomware doesn't wait.
Neither should your defense.

RANSHIELD — Detect. Block. Recover.